QP Code:25900329

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACCSCA00 - DIGITAL TECHNOLOGIES AND EMERGING TRENDS IN COMPUTING

(2024 ADMISSION ONWARDS)

Scheme for Evaluation

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Part A

One or two Sentences

Answer all questions

Each question carries 1 mark

1. **List any two types of Cloud Computing**
   Public Cloud, Private Cloud, Hybrid Cloud.
2. **Name one AI tool used in customer service**
   Chatbots (e.g., ChatGPT, IBM Watson Assistant, etc. or any other examples
3. **List two applications of Cloud Computing.**
   Data storage, Online backup, Software as a Service (SaaS), Email services. etc.
4. **List two applications of Augmented Reality**
   Gaming (e.g., Pokémon Go), Virtual Try-ons in shopping, medical visualization, Education.
5. **Describe cyber hygiene**
   Practices to maintain security and protect devices/data from cyber threats — like updating software, strong passwords, avoiding suspicious links.
6. **Write the full form of RPA.**
   Robotic Process Automation.
7. **List two types of robots.**
   Industrial robots, Service robots, Autonomous robots, Collaborative robots.
8. **Describe digital entrepreneurship?**
   Starting and managing a business using digital technologies and online platforms.
9. **Identify one research area where AI is applied.**
   Healthcare, Autonomous vehicles, Natural Language Processing, Robotics.
10. **Define digital banking**

Online or mobile-based banking that allows financial transactions via digital platforms.

[1x10 = 10]

Part B

Short Answers

Answer any 4 questions

Each question carries 5 marks

11. Define Artificial Intelligence and classify its main types.

AI refers to machines or systems that simulate human intelligence.

Types:

- Narrow AI (specific tasks) EG: Siri, Alexa, chatgpt
- General AI (human-level intelligence) eg: Hypothetical future AI
- Super AI (beyond human intelligence) eg: theoretical concept

12. **Explain the basic concepts of Cloud Computing**.

Cloud computing delivers computing services—servers, storage, databases, networking—over the internet.

Concepts:

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

13. Differentiate between threats, vulnerabilities, and risks with examples.

- *Threats:* A *potential danger* or harmful event that could exploit a weakness in a system. (e.g., Malware attacks (viruses, ransomware), Phishing emails, Unauthorized access, etc.)
- *Vulnerability:* A *weakness or flaw* in software, hardware, network, or human processes that can be exploited by a threat. (e.g., Outdated software, Weak passwords, Unpatched operating system, Unsecured Wi-Fi)
- *Risk:* potential damage from a threat exploiting a vulnerability (e.g., Data loss due to unpatched systems attacked by malware, financial loss from phishing scam)

14. Explain the components of VR.

Hardware: Head-Mounted Display, Controllers, Sensors
Software: VR applications and simulation programs
Content: 3D environment for interaction

15. Explain two major challenges faced by digital entrepreneurs.

Cybersecurity and data privacy issues
Market competition and technology changes
Funding and scalability challenges
Customer trust building

16. Compare the applications of AI in defence and education

Defense: autonomous drones, surveillance, predictive analytics etc
Education: personalized learning, grading automation, virtual tutors etc and explanation for each

[5x4 = 20]

Part C

Essay Questions
Answer any 2 questions
Each question carries 10 marks

17. **Explain the different service models (SaaS, PaaS, IaaS) with examples.**
    SaaS (Software as a Service): ready-to-use applications (e.g., Gmail, Office 365)
    PaaS (Platform as a Service): for app development (e.g., Google App Engine)
    IaaS (Infrastructure as a Service): virtual hardware resources (e.g., AWS EC2)
    Explanation of each in detail. (7.5 marks)
    Architecture diagram (2.5 marks)
18. **Explain the role of VR in remote learning.**
    Immersive learning experiences (virtual labs, 3D classrooms)
    Enhances engagement and understanding
    Bridges distance learning gaps (explanation: 7.5 marks)
    Examples: VR anatomy labs, virtual campus tours (2.5 marks)
19. **Discuss the architecture and components of a robot with a neat diagram.**
    Main Components:
    Sensors: detect environment
    Actuators: control motion
    Controller: brain of robot
    Power Supply
    End Effector: tool performing task (8 marks)
    Architecture Diagram: Input (Sensors) → Processor → Output (Actuators) (2 marks)

[10x2 = 20]